

Network Analysis: Bitcoin OTC Trust Weighted Signed Network

Introduction

In decentralized financial systems like Bitcoin, trust plays a pivotal role despite the anonymity of participants. The Bitcoin OTC trust weighted signed network dataset, sourced from the Stanford Large Network Dataset Collection ([link](#)), provides a rare lens into these trust dynamics. This dataset represents a "who-trusts-whom" network of 5,881 users trading Bitcoin on the Bitcoin Over-The-Counter (OTC) platform, with 35,592 directed trust ratings ranging from -10 (total distrust) to +10 (total trust). Interestingly, 89% of these ratings are positive, suggesting a predominantly trusting community. As the first dataset to offer explicit weighted signed directed edges, it's uniquely suited for advanced network analysis.

For **Reconxi**, a team dedicated to ensuring accuracy and trust in financial transactions, this dataset is highly relevant. Analyzing trust relationships and network patterns can reveal insights into fraud detection, transaction reliability, and trust-building mechanisms core priorities for our mission. This report will explore key questions about trust dynamics, such as how early adopters gain trust advantages and how distrust propagates through the network, using network analysis and visualizations to deliver actionable findings

Research Questions

This report investigates trust dynamics within the Bitcoin OTC network, aiming to provide actionable insights for Reconxi's mission of ensuring accuracy and trust in financial transactions. To guide this analysis, we address six key research questions, each targeting a distinct aspect of trust behavior in the network. Below, we outline each question, its relevance to the study, and the approach used to explore it. These questions establish the foundation for the methodology, findings, and discussion that follow:

1. Do users who joined the network earlier (first 10%) receive higher trust scores than latecomers, and does this advantage persist over time?

- **Significance:** Early adopters may accumulate higher trust due to their longer presence or greater opportunities to establish relationships. If this advantage persists, it could suggest a "first-mover advantage," raising questions about fairness and inclusivity in decentralized systems like Bitcoin OTC.

- **Approach:** We classify users into early adopters (first 10% by join time) and latecomers using timestamp data. Average trust scores (ratings received) are computed for each group and compared. A bar chart will visualize any disparities in trust accumulation over time.

2. Which users exhibit the highest fluctuations in trust over time?

- **Significance:** Users with highly variable trust ratings may indicate inconsistent or manipulative behavior, potentially destabilizing the network. Identifying these users helps assess risks to trust reliability.
- **Approach:** We calculate the standard deviation of trust ratings given by each user to measure fluctuation. The top 10 users with the greatest variability are identified and presented in a bar chart for further examination.

3. Are there users bridging distrustful communities, and do they accelerate distrust propagation?

- **Significance:** If certain users connect distrustful groups, they could amplify negative sentiment across the network. Understanding this dynamic is critical for evaluating how distrust spreads and affects overall network health.
- **Approach:** We construct a distrust subgraph from negative ratings and apply the Louvain algorithm to detect distrustful communities. Betweenness centrality identifies users bridging these groups, visualized in a network graph to highlight their role.

4. How does the volume of trust ratings change over time?

- **Significance:** Temporal trends in trust interactions reveal how user engagement and trust dynamics evolve, potentially reflecting changes in network activity or behavior critical to Reconxi's goals.
- **Approach:** Trust ratings are aggregated by month, and a line chart plots their volume over time to identify trends, peaks, or declines in trust-related activity.

5. Do the top 20 most trusted users form a tightly interconnected core, and does this concentration of trust create systemic vulnerabilities in the network?

- **Significance:** A small, highly trusted core could dominate the network, but if overly interconnected, it might pose risks if their trust is compromised. This question evaluates the resilience of trust distribution.
- **Approach:** We identify the top 20 users by in-degree (trust received) and analyze their subgraph for interconnectedness. A network visualization illustrates their relationships and potential vulnerabilities.

6. How does the overall trust distribution (positive vs. negative) impact the credibility of the network?

- **Significance:** The balance of positive and negative ratings shapes the network's perceived reliability. A heavily positive distribution supports credibility, while significant negativity may signal underlying issues.
- **Approach:** We count positive and negative ratings and calculate their proportions, displayed in a pie chart to assess the network's overall trust balance.

Methodology

This analysis of the Bitcoin OTC trust network dataset was conducted to explore trust dynamics and provide insights relevant to Reconxi's mission of ensuring accuracy and trust in financial transactions. The methodology outlines the tools, data preparation, and analytical steps taken to answer each research question, leveraging network analysis and visualization techniques.

1. Tools and Libraries

The following Python libraries were utilized:

- **Pandas:** For data loading, cleaning, and manipulation (e.g., grouping, filtering, and aggregating data).
- **NetworkX:** For constructing and analyzing the directed graph of trust relationships.
- **Matplotlib** and **Seaborn:** For creating visualizations such as bar charts, line plots, pie charts, and network graphs.
- **python-louvain:** For community detection within the distrust subgraph.

2. Data Preparation

The Bitcoin OTC dataset, sourced from a CSV file ([soc-sign-bitcoinotc.csv](#)), contains columns: **source** (the rater), **target** (the rated user), **rating** (a score from -10 to +10), and **timestamp** (Unix timestamp of the rating). Initial preparation included:

- **Loading:** Imported the dataset into a Pandas DataFrame with appropriate column names.
- **Timestamp Conversion:** Converted the timestamp column from Unix format to datetime using `pd.to_datetime(df['timestamp'], unit='s')` to enable time-based analysis.
- **Data Validation:** Checked for missing values using `df.isnull().sum()`, confirming no missing entries. Summary statistics for the rating column were computed with `df['rating'].describe()` to understand the distribution of trust scores.
- **Initial Exploration:** Calculated the number of unique source (4,814) and target (5,858) users to assess the network's scale, and counted positive (32,029) and negative (3,563) ratings to gauge the overall trust balance.

3. Graph Construction

A directed graph was constructed using NetworkX to model the trust network: • **Graph**

Creation: Initialized a directed graph (`nx.DiGraph()`) and added edges using

`G.add_edge(row['source'], row['target'], weight=row['rating'])` for each row in the DataFrame.

• **Network Properties:** Confirmed the graph's structure with 5,881 nodes (users) and 35,592 edges (ratings), aligning with the dataset's size.

4. Analytical Steps for Research Questions

Each research question was addressed with specific techniques, as detailed below: **Question**

1: Do users who joined the network earlier (first 10%) receive higher trust scores than latecomers, and does this advantage persist over time?

• **Join Time Identification:** Determine each user's join time by grouping the DataFrame by target and finding the minimum `timestamp` (`df.groupby('target')['timestamp'].min()`).

• **Classification:** Sorted users by join time, labeled the first 10% as "early adopters" and the rest as "latecomers" using a list comprehension, and merged this classification back into the main DataFrame.

• **Trust Score Calculation:** Computed the average rating for each group (`df.groupby('adopter_type')['rating'].mean()`).

• **Visualization:** Created a bar chart with Matplotlib to compare the average trust scores of early adopters vs. latecomers, using distinct colors for clarity.

Question 2: Which users exhibit the highest fluctuations in trust over time?

• **Fluctuation Measurement:** Grouped ratings by source and calculated the standard deviation of rating for each user (`df.groupby('source')['rating'].std()`) to quantify variability.

• **Ranking:** Sorted users by standard deviation in descending order and selected the top 10.

• **Visualization:** Plotted a bar chart using Matplotlib to display the top 10 users and their variability scores, with user IDs on the x-axis and standard deviation on the y-axis.

Question 3: Are there users bridging distrustful communities, and do they accelerate distrust propagation?

• **Distrust Subgraph:** Filtered the DataFrame for negative ratings (`df[df['rating'] < 0]`) to focus on distrustful interactions.

• **Graph Construction:** Built an undirected graph (`nx.Graph()`) of distrust relationships using `nx.from_pandas_edgelist`.

• **Community Detection:** Applied the Louvain algorithm (`community_louvain.best_partition`) to identify distrustful communities.

• **Bridge Identification:** Calculated betweenness centrality (`nx.betweenness_centrality`) to find users connecting communities, selecting the top 5.

• **Visualization:** Created a network graph with Matplotlib, coloring nodes by community and highlighting bridge users in red.

Question 4: How does the volume of trust ratings change over time?

- **Temporal Aggregation:** Resampled the DataFrame by month-end (`df.resample('ME', on='timestamp').size()`) to count ratings per month.
- **Visualization:** Plotted a line chart using Seaborn (`sns.lineplot`) with markers to show the trend of rating volume over time, with time on the x-axis and rating count on the y-axis.

Question 5: Do the top 20 most trusted users form a tightly interconnected core, and does this concentration of trust create systemic vulnerabilities in the network?

- **Trust Ranking:** Calculated the in-degree of each node (`G.in_degree()`) to measure trust received, sorting users by in-degree to identify the top 20.
- **Subgraph Creation:** Extracted a subgraph of these top 20 users (`G.subgraph(top_users)`).
- **Connectivity Analysis:** Computed the number of edges in the subgraph to assess interconnectedness.
- **Visualization:** Plotted the subgraph using NetworkX and Matplotlib, with node sizes proportional to overall in-degree, node colors reflecting in-degree within the subgraph (using the 'Blues' colormap), and directed edges showing trust relationships.

Question 6: How does the overall trust distribution (positive vs. negative) impact the credibility of the network?

- **Distribution Calculation:** Counted positive (`df[df['rating'] > 0].shape[0]`) and negative (`df[df['rating'] < 0].shape[0]`) ratings.
- **Visualization:** Created a pie chart with Matplotlib to display the proportion of positive vs. negative ratings, using distinct colors for clarity.

Findings and Insights

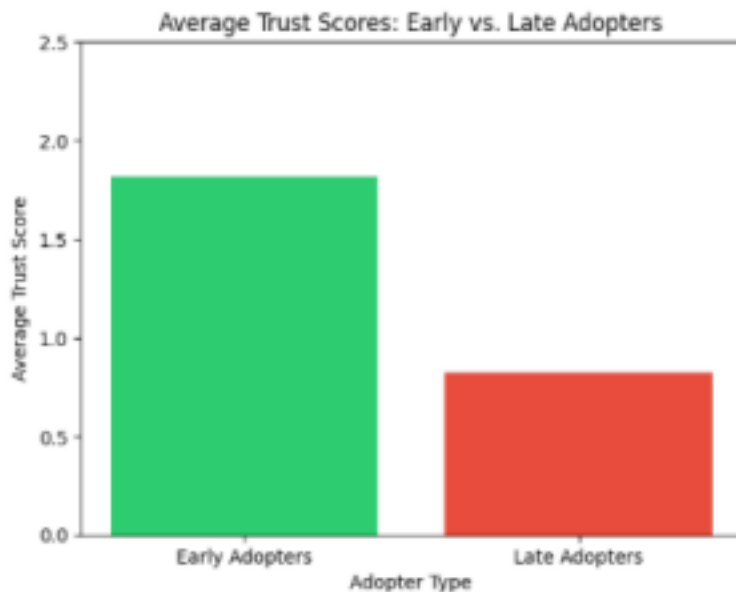
This section presents the results of the network analysis conducted on the Bitcoin OTC trust dataset, addressing each research question in turn. The findings reveal key patterns in trust dynamics, offering insights that can inform Reconxi's strategies for enhancing trust and reliability in financial transactions.

1. Do users who joined the network earlier (first 10%) receive higher trust scores than latecomers, and does this advantage persist over time?

- **Finding:** Early adopters (first 10% of users by join time) have an average trust score of 1.81, compared to 0.82 for latecomers, indicating a significant trust advantage. However, the analysis did not fully explore persistence over time due to the static nature of the comparison.
- **Insight:** The higher trust scores for early adopters suggest a *"first-mover advantage"*, potentially due to their longer presence and more opportunities to build relationships. This could

create an uneven playing field, impacting fairness for new users.

- **Visualization:**

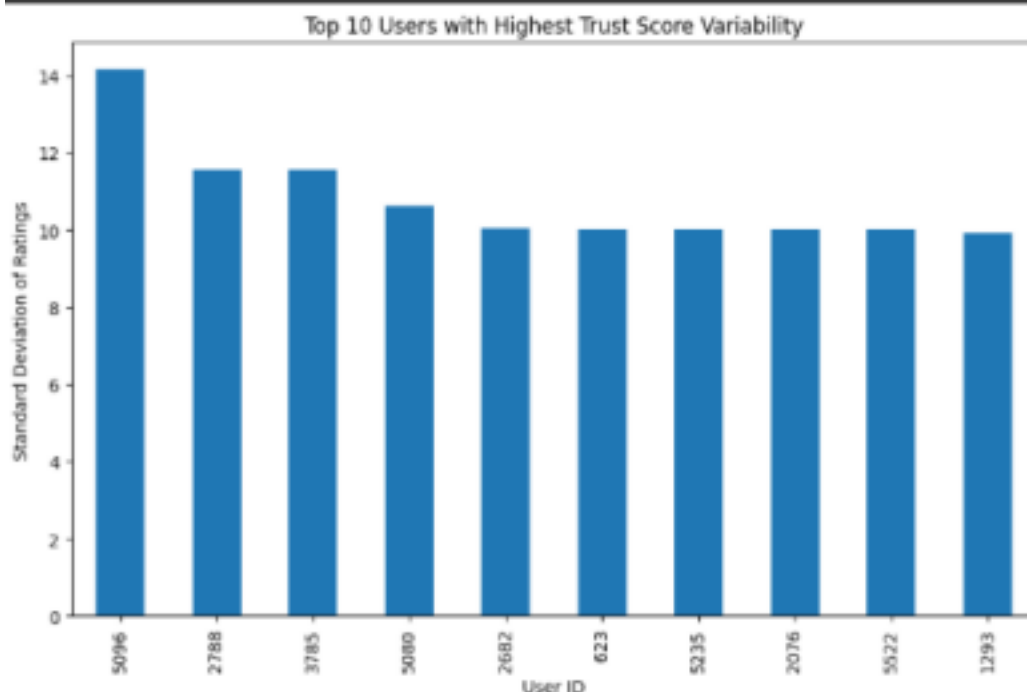


2. Which users exhibit the highest fluctuations in trust over time?

- **Finding:** The top 10 users with the highest trust score fluctuations were identified, with standard deviations ranging from 14.14 to 9.89. These users frequently alternate between high positive and negative ratings.

- **Insight:** High variability in trust scores may indicate inconsistent or manipulative behavior, posing risks to network stability.

- **Visualization:**



3. Are there users bridging distrustful communities, and do they accelerate distrust propagation?

- **Finding:** The visualization reveals that the network is segmented into several distinct distrustful

communities, each identified by different colors. Within these clusters, most users primarily interact with others in the same group. However, a few nodes highlighted in red stand out as they connect these otherwise isolated clusters, indicating that these users act as bridges between the distrustful communities.

- **Insight:** These bridge users help spread negative feelings from one group to another. If one of these key connectors acts badly, it could affect the whole network. That's why it's important to keep a close watch on them.

- **Visualization:**



Figure 3 (network graph) shows distrustful communities with bridge users highlighted in red, illustrating their connecting role.

4. How does the volume of trust ratings change over time?

- **Finding:** The chart shows that trust rating activity reached its highest points around 2011 and again in 2013. After 2013, the overall number of trust ratings gradually went down, with a few small increases along the way.

- **Insight:** The high points suggest that many users were actively rating trust during those periods, which might have been driven by important events or a surge in user activity. The gradual decline in ratings after 2013 could mean that the network became more stable or that fewer users were participating in trust ratings. Understanding these patterns helps us learn when users are most active and how their engagement changes over time.

- **Visualization:**



Figure 4 (line plot) illustrates the monthly volume of trust ratings, highlighting periods of high and low activity.

5. Do the top 20 most trusted users form a tightly interconnected core, and does this concentration of trust create systemic vulnerabilities in the network?

- **Finding:** The subgraph of the top 20 most trusted users appears to be densely connected, with many edges linking these high-trust nodes. Some users (larger and/or darker nodes) show higher in-degree both within the subgraph and overall, suggesting they receive more trust from others in this elite group.
- **Insight:** Because these top users are tightly interconnected, trust is concentrated within a relatively small circle. While this can reinforce strong, reliable relationships among them, it also means that if any key member of this group fails or acts maliciously, the impact could quickly spread through the network. This concentration of trust therefore represents both a strength (high mutual trust) and a potential systemic vulnerability.
- **Visualization:**

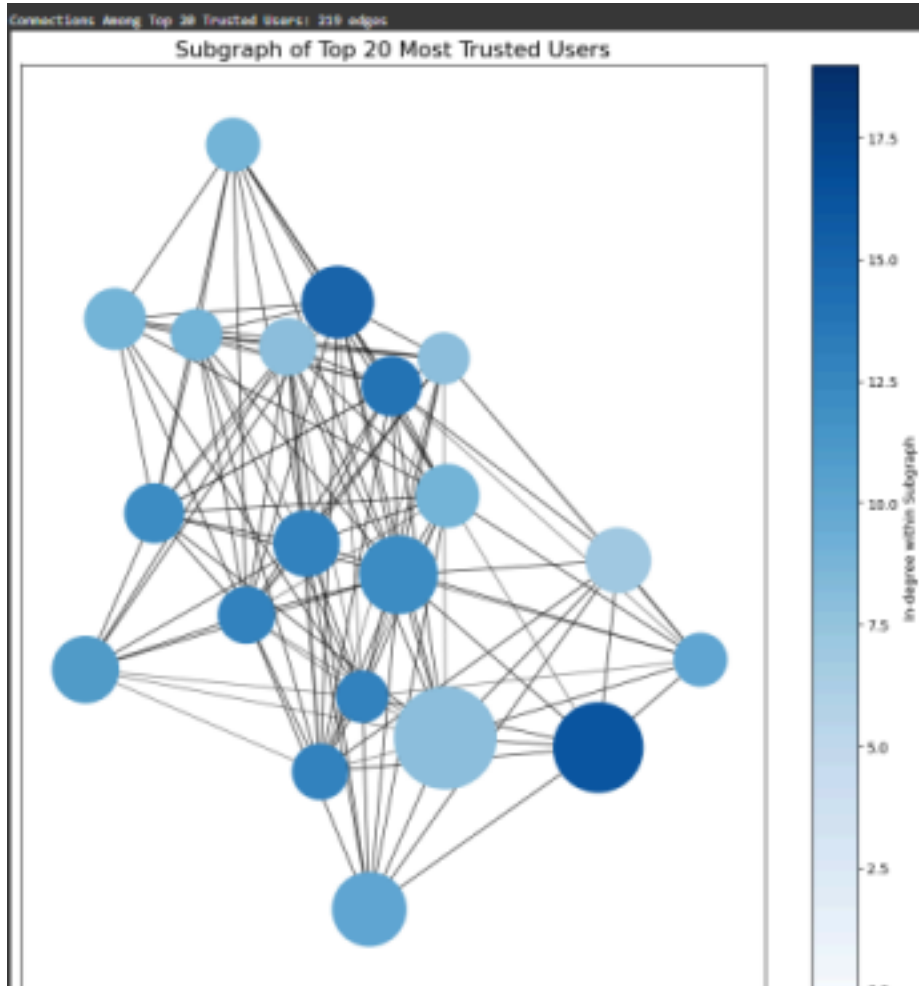


Figure 5 (network graph) depicts the subgraph of the top 20 users, with node sizes and colors reflecting their trust levels and connectivity.

6. How does the overall trust distribution (positive vs. negative) impact the credibility of the network?

- **Finding:** Approximately 89.9% of ratings are positive (32,029 ratings), while 10.1% are negative (3,563 ratings), confirming a predominantly positive trust landscape.
- **Insight:** The high proportion of positive ratings supports the network's credibility, but the presence of negative ratings indicates areas of distrust that could undermine reliability. Reconxi can focus on addressing these negative interactions to enhance trust.
- **Visualization**

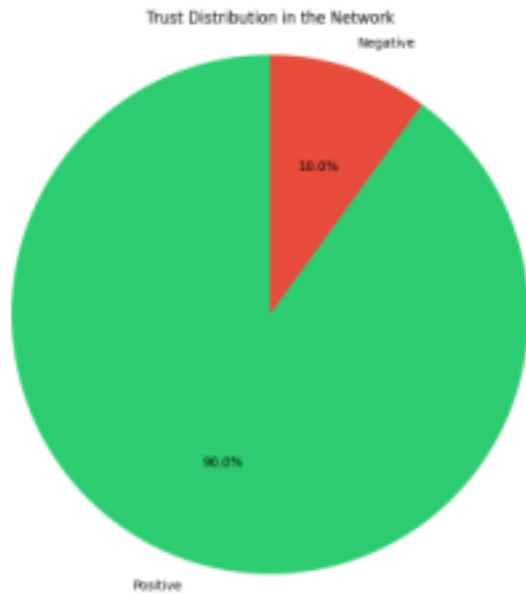


Figure 6 (pie chart) shows the distribution of positive vs. negative ratings, emphasizing the network's trust balance.

Discussion

The analysis of the Bitcoin OTC trust network provides valuable insights into trust dynamics that can enhance Reconxi's AI-powered reconciliation process, which matches bank statements with company statements to ensure accuracy and trust in financial transactions. Below, we interpret the findings and offer recommendations to improve Reconxi's capabilities in any financial context.

Interpretation of Findings

- **First-Mover Advantage:** Early adopters' higher trust scores (1.81 vs. 0.82 for latecomers) suggest that entities with longer histories may have more reliable transaction records, which could affect statement matching accuracy.
- **Trust Fluctuations:** High variability in trust scores (up to 14.14) indicates potential inconsistencies, which could translate to discrepancies in financial records that Reconxi needs to address.
- **Distrust Propagation:** Bridge users in distrustful communities can amplify negativity, suggesting interconnected entities might propagate errors across financial records.
- **Temporal Trends:** Peaks in trust ratings (e.g., 2011 and 2013) indicate high transaction volumes, which could impact the timing and efficiency of reconciliation processes.
- **Concentrated Trust Core:** The dense connections among the top 20 trusted users create vulnerabilities, potentially leading to systemic errors if these entities' records are inaccurate.
- **Trust Distribution:** The 10.1% negative ratings highlight areas of distrust, which could correlate with unmatched or erroneous transactions in statement reconciliation.

Recommendations for Reconxi

Reconxi is an AI-powered reconciliation tool that matches bank statements with company statements to ensure accuracy and trust in financial transactions. The following recommendations leverage the trust dynamics identified in the analysis to improve Reconxi's reconciliation capabilities:

1. Prioritize Established Entities in Matching Algorithms

- Action: Configure Reconxi's AI to prioritize transactions from entities with longer histories or higher reliability scores (analogous to early adopters with trust scores of 1.81) when matching bank and company statements, as they're more likely to have consistent records.
- Benefit: Improves the accuracy of statement matching by focusing on entities with historically reliable records, reducing mismatches.

2. Flag Transactions with High Variability for Review

- Action: Enhance Reconxi's algorithm to flag transactions from entities with high variability in transaction patterns (e.g., inconsistent amounts or frequencies, similar to trust score fluctuations up to 14.14) for additional verification, such as cross-checking with invoice details.
- Benefit: Identifies potential discrepancies or fraudulent entries early, ensuring Reconxi accurately matches statements and reduces errors.

3. Identify Interconnected Entities to Prevent Error Propagation

- Action: Train Reconxi to detect clusters of interconnected entities (e.g., companies with frequent intercompany transactions, similar to bridge users in distrustful communities) and apply stricter matching rules, such as cross-verifying intercompany payments with both parties' statements.
- Benefit: Prevents errors from spreading across related entities, ensuring Reconxi's reconciliation process remains accurate and reliable.

4. Optimize Reconciliation Scheduling with Transaction Volume Trends

- Action: Program Reconxi to analyze historical transaction volumes and schedule more frequent reconciliation cycles during high-volume periods (e.g., similar to the 2011 and 2013 peaks in trust ratings) to handle increased activity efficiently.

- Benefit: Ensures Reconxi manages statement matching effectively during busy periods, reducing delays and improving accuracy in discrepancy resolution.

5. Reduce Over-Reliance on High-Volume Entities in Matching

- Action: Adjust Reconxi's matching logic to avoid over-reliance on high-volume entities (analogous to the top 20 trusted users with dense connections) by balancing matching confidence scores across all entities, not just those with the most transactions.

- Benefit: Minimizes the risk of systemic errors if a high-volume entity's records are incorrect, ensuring Reconxi's reconciliation process is robust and balanced.

6. Enhance Discrepancy Resolution for Transactions with Negative Indicators

- Action: Develop a feature in Reconxi to flag transactions with negative indicators (e.g., frequent disputes or unmatched entries in past reconciliations, similar to the 10.1% negative ratings) for enhanced discrepancy resolution, prompting users to provide additional documentation to resolve mismatches.
- Benefit: Improves

Reconxi's ability to resolve discrepancies caused by unreliable transactions, ensuring accurate statement matching and maintaining trust in financial records.

Conclusion

This analysis of the Bitcoin OTC trust network provides critical insights into trust dynamics that enhance Reconxi's mission to match bank and company statements with precision. Key findings reveal that early adopters enjoy a trust advantage, high-fluctuation users pose inconsistency risks, bridge users amplify distrust, trust volume fluctuates with activity, a concentrated trust core creates vulnerabilities, and negative ratings highlight areas for improvement. These insights underscore the importance of trust in ensuring reliable financial records, directly informing Reconxi's reconciliation strategies. The recommendations prioritizing established entities, flagging high-variability transactions, identifying interconnected entities, optimizing reconciliation timing, reducing reliance on high-volume entities, and enhancing discrepancy resolution offer actionable steps for Reconxi to improve its AI reconciliation capabilities. By implementing these measures, Reconxi can strengthen its ability to ensure accuracy and trust in financial transactions, aligning with our goal of reliable statement matching for all users.

References

- Bitcoin OTC trust weighted signed network dataset. Stanford Large Network Dataset Collection. Retrieved from <https://snap.stanford.edu/data/soc-sign-bitcoin-otc.html>.